

Skills Summary

Generating and Comparing Two Patterns

Use this reference while completing your worksheet or when studying.

Skill 1 — Generating a term without listing them all

Rule: The first term is the start, so reaching term n takes $n - 1$ steps.

term $n = \text{start} + \text{step size} \times (n - 1)$

Off-by-one is the usual slip here: term 6 is 5 steps out, not 6.

Example: A pattern starts at 3 and adds 5 each time. Write the 7th term.

Step 1. Listing seven terms works but invites a miscount, so I count steps instead: the 7th term is six steps past the start.

Step 2. $3 + 5 \times 6 = 3 + 30$, and listing agrees: 3, 8, 13, 18, 23, 28, 33. 7th term: **33**

Try it: A pattern starts at 2 and adds 6 each time. Write the 5th term.

check: 29

Skill 2 — Relating one rule to the other

Rule: When two patterns start at the same place, compare their *step sizes*, not their terms.

If both start at 0 and one step is k times the other, then every matching term is k times the other as well — after the same number of steps, one pattern has added k times as much.

This fails if the two patterns start at different places, because the head start never gets multiplied.

Example: Pattern C starts at 0 and adds 3. Pattern D starts at 0 and adds 9. Each D term is how many times the matching C term?

Step 1. Both start at 0, so the whole relationship is carried by the steps and I can compare the step sizes alone.

Step 2. $9 \div 3 = 3$, and checking term 4: C gives 9 and D gives 27. Multiplier: **3**

Try it: One pattern starts at 0 and adds 5; another starts at 0 and adds 20. Each term of the second is how many times the first?

check: 4

Skill 3 — Comparing matching terms

Rule: To compare term n of two patterns, generate both and subtract.

gap = $(\text{start}_1 + \text{step}_1 \times (n - 1)) - (\text{start}_2 + \text{step}_2 \times (n - 1))$

A pattern that begins behind can still finish ahead: the gap closes by $(\text{step}_1 - \text{step}_2)$ every step.

Example: Pattern E starts at 10 and adds 2. Pattern F starts at 4 and adds 4. How much greater is F's 8th term than E's 8th term?

Step 1. The starts differ, so a step-size ratio will not answer this — I build both 8th terms and subtract.

Step 2. F: $4 + 4 \times 7 = 32$. E: $10 + 2 \times 7 = 24$. Gap: $32 - 24$. F is greater by 8

Try it: Pattern G starts at 6 and adds 3; pattern H starts at 1 and adds 5. How much greater is H's 9th term than G's 9th term?

check: 11

Skill 4 — Writing the two patterns as ordered pairs

Rule: Pair the terms that share a term number: (term n of the first pattern, term n of the second).

The pattern named first supplies the across number. Plotted on a grid, matching terms of two constant-step patterns always fall on a straight line through the pairs.

Example: Pattern J starts at 0 and adds 2. Pattern K starts at 0 and adds 5. Write the ordered pair for term 6.

Step 1. The question names J first, so J's term is the across number and K's is the up number — I generate each to term 6 separately.

Step 2. J: $0 + 2 \times 5 = 10$. K: $0 + 5 \times 5 = 25$. Ordered pair: **(10, 25)**

Try it: Pattern L starts at 0 and adds 3; pattern M starts at 0 and adds 7. Write the ordered pair for term 5.

check: (12, 28)